

Interpretable Time Series Autoregression for Periodicity Quantification

Complex, dynamically evolving systems such as urban transportation and climate generate many related time series data. Monitoring their periodic patterns can guide operations and policy by revealing regular behavior, disruptions, and recovery. However, quantifying these patterns is challenging: only a few lags are typically meaningful, observations are noisy and affected by shocks, periodicity varies across time and location, and the resulting datasets can contain millions of series. We develop an optimization framework for interpretable autoregression that addresses these challenges directly. Sparse nonnegative coefficients identify dominant periodic lags, grouped constraints encode calendar structure, robust losses limit the influence of disruptions, and stable lag vocabularies enable consistent comparisons across related series. We first formulate the full model as an exact mixed-integer optimization problem, then develop a sequence of increasingly scalable algorithms: iteratively reweighted least squares replaces the robust objective with weighted quadratic subproblems; decision-variable pruning screens candidate lags before solving a reduced mixed-integer problem; and a global-local procedure learns a common lag vocabulary before estimating local coefficients independently. These accelerated algorithms compute high-quality solutions while reducing runtime by more than two orders of magnitude, scaling the framework to more than three million time series in under a minute. On airport ridehailing data, the interpretable model exceeds the predictive accuracy of dense autoregression while using a substantially smaller, more interpretable set of lags and robust objectives. Across case studies spanning Manhattan ridehailing and subway ridership, the framework quantifies weekly periodicity consistently across time and location, allowing to analyze the pandemic recovery and the effect of policy intervention. The empirical studies on North American climate and global sea surface show the long-term evolution of dynamic systems with respect to yearly seasonality, while remaining tractable at scale.

Key words: Interpretable machine learning, time series analysis, periodicity quantification, sparse autoregression, human mobility, climate systems

1. Introduction

Urban transportation systems are organized around recurring activity: passenger flows rise and fall with morning and evening commutes within urban areas, airport trips follow flight schedules, weekday and weekend travels differ systematically due to different trip purposes, and seasonal conditions affect travel mode use (Gonzalez et al. 2008, Song et al. 2010). These periodic patterns are operationally useful because they indicate when travel demand is predictable, where capacity should be allocated, how quickly a system returns to routine after a disruption, and how a policy intervention affects urban mobility. They are also not

static: regularity can weaken during shocks and policy interventions, recover at different speeds across neighborhoods, and differ substantially across travel modes. Thus, quantifying this regularity and tracking how it evolves therefore requires data-driven models that can adapt to time-varying system behavior.

Figure 1 makes this idea concrete using mobility data from Manhattan in New York City, USA. Denote by $\{x_t\}$ the time series of hourly trip counts. We say that lag k matches a periodic cycle if x_t and x_{t-k} tend to have similar magnitudes at lag $k \in \mathbb{Z}^+$ throughout the evolution of the time series. In this example, we show three hourly trip time series with a visible weekly rhythm. At the weekly lag $k = 168$, the points $\{(x_{t-168}, x_t)\}$ concentrate near the diagonal line, indicating tight alignment. We also show how the same idea extends to a system-level periodicity measure of ridehailing trips across 66 areas in Manhattan. Although trip counts alone do not determine how regular an area is, the learned weekly autocorrelations quantify how strongly each area’s trips recur from week to week. This example validates an important interpretation: areas with larger weekly autocorrelations show tighter x_t -versus- x_{t-168} alignment and more consistent profiles across consecutive weeks.

The same methodological problem for periodicity quantification appears beyond urban transportation systems (Kutz et al. 2016, Piccardi et al. 2024). For example, climate systems exhibit periodic and seasonal patterns in temperature, precipitation, ocean circulation, and atmospheric dynamics, and this seasonality evolves over time due to long-term variability and climate change. Accurately characterizing these spatiotemporal patterns and monitoring how they change are essential for understanding climate dynamics, anticipating extreme events, managing agriculture production, and supporting policy-making. These observations motivate the central question of the paper:

Research Question. *How can we learn a small set of calendar-relevant lags—local, daily, weekly, and yearly—that explain recurring behavior in time series, estimate how strongly those lags appear across locations, time segments, and system components, and do so reliably under disruptions and at scale?*

Our starting point is the classical autoregression (AR) model (Brockwell and Davis 1991). For a univariate time series $x = (x_1, \dots, x_T)^\top \in \mathbb{R}^T$ of T time steps, an order- d AR model represents each observation as a linear combination of its past values as

$$x_t = \sum_{k=1}^d w_k x_{t-k} + \varepsilon_t, \quad t = d+1, \dots, T.$$

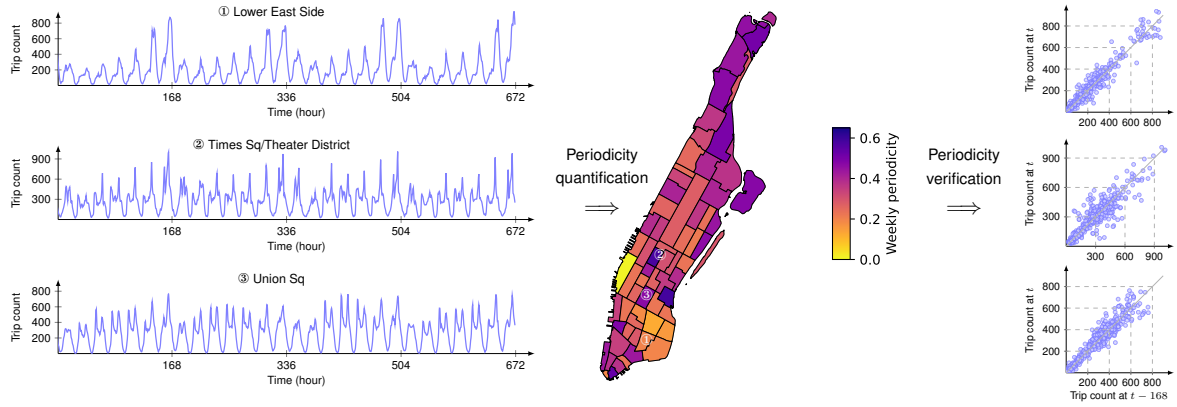


Figure 1 Motivating examples of periodicity in hourly ridehailing dropoff trip time series. The left panel shows hourly dropoff trip time series for three areas in Manhattan (① Lower East Side, ② Times Sq/Theater District, ③ Union Sq) from January 5, 2025 to February 1, 2025. The middle panel maps the weekly periodicity of 66 areas in Manhattan, with the three example areas marked. The right panel illustrates lag dependence by plotting the current trip count x_t against the lagged trip count x_{t-168} for a weekly cycle. Stronger recurrence at lag $k = 168$ appears as tighter alignment around the diagonal line.

We say that the series has positive periodicity at lag q over an estimation window if x_t and x_{t-q} tend to move together: high values of the series are followed, one cycle later, by high values, and low values are followed by low values. If we define the lag operator L such that $L^k x_t = x_{t-k}$, strict periodicity means that $\{x_t\}$ lies in the null space of the operator $(I - L^q)$, i.e., $(I - L^q)x_t = 0, \forall t$. For centered data, this alignment is measured by the lag- q sample autocorrelation:

$$\hat{\rho}(q) = \frac{\sum_{t=q+1}^T x_t x_{t-q}}{\left(\sum_{t=q+1}^T x_t^2\right)^{1/2} \left(\sum_{t=q+1}^T x_{t-q}^2\right)^{1/2}}.$$

A positive value of $\hat{\rho}(q)$ is the statistical signature of recurrence at lag q ; by contrast, negative lag dependence describes alternation rather than repeated activity. If we fit the one-lag AR model $x_t = w_q x_{t-q} + \varepsilon_t$, least squares gives $\hat{w}_q = \sum_{t=q+1}^T x_t x_{t-q} / \sum_{t=q+1}^T x_{t-q}^2$. This is the unnormalized version of $\hat{\rho}(q)$; the two coincide when the lagged series are standardized over the estimation window. With multiple selected lags $\Omega \subseteq [d]$, the coefficients are recovered from the lag-covariance normal equations $(A_\Omega^\top A_\Omega) \hat{w}_\Omega = A_\Omega^\top \tilde{x}$, where A_Ω is the lagged design matrix restricted to Ω . Specifically, $A_\Omega = \{\tilde{x}_k\}_{k \in \Omega} \in \mathbb{R}^{(T-d) \times |\Omega|}$ where $\tilde{x}_k = (x_{d+1-k}, \dots, x_{T-k})^\top \in \mathbb{R}^{T-d}$. The target vector is $\tilde{x} = (x_{d+1}, \dots, x_T)^\top \in \mathbb{R}^{T-d}$. Thus, an AR coefficient at lag q measures conditional lag dependence after accounting for the other

selected lags. In the periodicity quantification model, we call $\Omega = \{k : \hat{w}_k > 0\}$ the periodicity signature and interpret $\hat{w}_q$ as the periodicity strength at calendar lag q . In the case of hourly time series data, $q = 24$ and $q = 168$ correspond to daily and weekly recurrence; for monthly time series data, $q = 12$ corresponds to yearly seasonality.

Classical AR is not sufficient to answer our research question, which raises five challenges. First, *dominance*: dense AR coefficients may predict well but do not isolate the small set of lags that explain recurring behavior. Second, *sign*: signed coefficients can mix positive recurrence with compensating negative effects, whereas periodicity discovery requires positive self-similarity. Third, *calendar tolerance*: recurring behavior may appear near, rather than exactly at, canonical lags such as 24 or 168. Fourth, *disruption*: holidays, weather, strikes, airport irregularities, policy interventions, and shocks such as COVID-19 can distort squared-loss estimates. Fifth, *stability and scale*: citywide, multimodal, and climate panels can contain millions of related time series and require a common calendar-lag vocabulary so that coefficient changes across locations, time segments, and system components have the same interpretation.

To address these challenges, we develop a unified optimization framework for interpretable periodicity quantification. The framework modifies AR so that the selected lag structure is sparse, positive, robust, and stable across linked time series, while the accompanying algorithms scale to large panels. Across our case studies, this framework recovers substantially sparser lag supports than dense AR without sacrificing predictive accuracy, our accelerated algorithms reduce runtime by more than two orders of magnitude relative to exact optimization, and the resulting framework scales to more than three million time series in under a minute. Our contributions are three-fold.

- **Modeling.** We formulate periodicity quantification as sparse nonnegative AR. The selected support defines a periodicity signature, and the corresponding coefficients quantify the strength of recurrence at local, daily, weekly, or yearly lags. We then add grouped calendar-lag constraints to account for near-calendar aliases, robust losses to reduce the influence of disruptions, and support-stability constraints to track the same calendar-lag vocabulary across locations, time segments, and system components. This yields a single model family for sparsity, calendar tolerance, robustness, and stability across time and space.

- **Algorithms.** We develop optimization algorithms that match the structure of the model. Exact mixed-integer quadratic optimization (MIQP) selects sparse lag supports when support quality is central to interpretation. Grouped mixed-integer constraints encode calendar windows without forcing the model to choose a single prespecified lag. Slack and iteratively reweighted formulations handle Huber robustness. Decision-variable pruning (DVP) makes stable multi-segment support-selection problems tractable by screening candidate lags before solving the final coupled mixed-integer optimization (MIO) problem. Finally, global-support/local-estimation decompositions make large spatial panels tractable by learning a common calendar-lag vocabulary and then estimating local coefficient strengths.

- **Empirical evidence.** We organize the empirical analysis as a sequence of increasingly complex case studies. In airport ridehailing data, exact support selection and robust losses are tested in a simple temporal setting where disruption can change both lag selection and prediction quality. In citywide ridehailing and subway data, the weekly coefficient tracks the disruption and recovery of mobility regularity across years and neighborhoods. In Manhattan ridehailing data, the same lag vocabulary compares recurrence across phases of a congestion pricing policy. Finally, North American climate and global sea surface temperature data test whether the same framework scales to large panels with yearly seasonality and produces interpretable spatial seasonality heatmaps beyond urban mobility.

The remainder of the paper is organized as follows. Section 2 summarizes the literature related to time series modeling, sparse regression, interpretable machine learning, and optimization-based support selection. Section 3 develops the periodicity quantification model, starting from sparse nonnegative AR and adding grouped calendar lags, robustness, and stability across segments. Section 4 presents the corresponding optimization algorithms. Section 5 reports the empirical case studies. Section 6 concludes this study with limitations and extensions.

2. Related Literature

Our work lies at the intersection of autoregressive time series modeling, sparse and robust regression, optimization-based variable selection, and interpretable spatiotemporal analysis. We review these streams selectively, emphasizing the distinction between forecasting or latent-pattern recovery and our objective of quantifying recurrence through a small, stable set of calendar-relevant lags.

Autoregression and interpretable spatiotemporal dynamics. Time series models are foundational in transportation, climate science, econometrics, and dynamical systems analysis (Kutz et al. 2016, Cochrane 1997). Classical autoregressive, moving-average, and ARIMA models represent serial dependence and seasonality through lagged observations and shocks (Brockwell and Davis 1991). Time-varying AR and VAR models allow coefficients to evolve across windows or regimes (Primiceri 2005, Bringmann et al. 2017, Haslbeck et al. 2021); low-rank parameterizations further compress and interpret high-dimensional time-varying dynamics (Harris et al. 2021, Chen et al. 2024). These models characterize changing system dynamics, whereas our primary estimand is the strength of positive recurrence at calendar lags under a stable interpretation across segments.

High-order AR and VAR models can obscure the dominant lag relationships. Sparse AR and sparse VAR address this issue through regularization or explicit support restrictions (Nardi and Rinaldo 2011, Tibshirani 1996, Basu and Michailidis 2015, Davis et al. 2016, Carrizosa et al. 2017). Structured regression also permits sparse models to vary over temporal or spatial graphs (Bertsimas et al. 2024). Our approach is more specialized: each feature is a past observation, nonnegative selected coefficients have a recurrence interpretation, and grouped lag windows encode calendar aliases such as 23/24 or 167/168.

A closely related recent approach learns sparse nonnegative convolutional kernels to reveal local and cyclical correlations in time series (Chen et al. 2025a). Our framework instead uses one-sided autoregression, so a coefficient represents dependence on a past lag conditional on the other selected lags; it also adds exact structured support selection, robust loss, and a common lag vocabulary across linked segments. In transportation, recent work has used tensor factorization, interpretable nonnegative decomposition, and nonstationary temporal matrix factorization to identify mobility patterns or improve traffic forecasts (Du et al. 2019, Li et al. 2023, Chen et al. 2025b). These methods motivate interpretable and scalable spatiotemporal modeling, but they target latent-factor recovery, data completion, or prediction rather than a direct coefficient-based measure of daily, weekly, or yearly recurrence.

Exact and structured sparse regression. Sparse regression has traditionally been approached through greedy methods such as orthogonal matching pursuit, CoSaMP, and subspace pursuit (Pati et al. 1993, Needell and Tropp 2009, Dai and Milenkovic 2009); convex

regularization such as the LASSO (Tibshirani 1996); folded-concave and other nonconvex penalties (Fan et al. 2011); and Bayesian variable-selection methods that quantify uncertainty in the selected variables (Wang et al. 2020). These approaches can be computationally attractive, but a penalized or heuristic solution need not satisfy a prescribed support cardinality or cross-segment support relationship exactly. Modern MIO techniques have made best-subset selection practical for substantially larger regression problems and provides optimality certificates for the formulated support-selection problem (Bertsimas et al. 2016, Bertsimas and Van Parys 2020, Hazimeh et al. 2022). Subsequent work develops alternative exact formulations, fast local-combinatorial algorithms, shrinkage-enhanced subset selection, and stronger convexifications (Gómez and Prokopyev 2021, Hazimeh and Mazumder 2020, Mazumder et al. 2022, Atamturk and Gomez 2025).

Our contribution builds on the aforementioned literature but changes both the estimand and the model structure. We do not treat lag selection as generic feature selection: selected nonnegative lags define a periodicity signature, coefficient mass around a calendar window defines recurrence strength, and a shared vocabulary gives coefficient changes the same meaning across periods and locations. The exact MIO formulation incorporates these restrictions together with robustness and support stability; DVP retains a final coupled MIO on a screened lag set, and global–local estimation provides a distinct scalable decomposition for very large panels. Altogether, the paper integrates established components from sparse AR, robust estimation, and exact subset selection around the specific problem of interpretable periodicity quantification.

Robust regression. Robust regression limits the influence of atypical observations and heavy-tailed noise. Huber’s M-estimator combines a quadratic loss near zero with linear tails and admits convenient convex optimization representations (Huber and Ronchetti 2009, Boyd and Vandenberghe 2004). Quantile regression provides an asymmetric absolute-loss alternative, and conformalized quantile regression uses quantile estimates to construct calibrated prediction intervals (Koenker and Hallock 2001, Romano et al. 2019). More generally, fast algorithms such as iteratively reweighted least squares have been developed for ℓ_p -regression, including convergence analyses for nonsmooth regression and related sparse-recovery problems (Adil et al. 2019, 2024, Kümmerle et al. 2021, Peng et al. 2022). Recent optimization work also studies scalable risk-based sparse regression and exact outlier selection through conic mixed-integer formulations (Wu et al. 2023, Gómez and Neto 2025).

We leverage robustness for a narrower purpose: preventing disruptions and isolated abnormal periods from determining the selected periodicity signature. The robust model is not intended as a causal change-point or outlier-classification procedure.

3. Periodicity Quantification with Interpretable AR

The introduction connected periodicity in a single time series to positive lag dependence and AR coefficients. We now extend that idea to collections of related time series and develop our general framework for interpretable periodicity quantification, which is the goal of this section.

Let $\mathcal{S}$ denote a set of *series segments*. A segment $s \in \mathcal{S}$ is the object on which an AR coefficient vector is estimated. Depending on the application, a segment may correspond to a station, an airport zone, a neighborhood-year pair, a location-mode pair, a location-period pair, or a climate grid-cell period. We write $x_s = (x_{s,1}, \dots, x_{s,T_s})^\top \in \mathbb{R}^{T_s}$ for the observed time series in segment s . The notation reflects the preprocessing used for a particular empirical comparison; e.g., centering gives the autocorrelation interpretation described in Section 1, and we hold any scaling or normalization used in experiments fixed within that comparison.

For an AR model of order $d \in \mathbb{Z}^+$, we define the estimation window $\mathcal{T}_s = \{d+1, \dots, T_s\}$. For each s , we let $A_s \in \mathbb{R}^{|\mathcal{T}_s| \times d}$ be the lagged design matrix whose row associated with time $t \in \mathcal{T}_s$ is $(x_{s,t-1}, x_{s,t-2}, \dots, x_{s,t-d})$, and let $\tilde{x}_s = (x_{s,d+1}, \dots, x_{s,T_s})^\top \in \mathbb{R}^{|\mathcal{T}_s|}$. For a coefficient vector $w_s \in \mathbb{R}^d$, the residual at time t is $\varepsilon_{s,t}(w_s) = x_{s,t} - \sum_{k=1}^d w_{s,k} x_{s,t-k}$. The goal is to estimate w_s by minimizing a chosen loss function, while learning an interpretable lag structure. We therefore build the model around three objects: the periodicity signature, the periodicity strength, and the calendar-lag vocabulary:

- The selected support of segment s is $\Omega_s = \text{supp}(w_s) = \{k \in [d] : w_{s,k} \neq 0\}$. We call Ω_s the *periodicity signature* of segment s . It identifies which lags are used to explain recurring behavior.

- A coefficient $w_{s,q}$ at a reference lag q is interpreted as a *periodicity strength* when it is nonnegative. For hourly mobility data, the most important reference lags are typically $q = 1, 24, 168$, corresponding to local dependence, daily recurrence, and weekly recurrence, respectively. For monthly climate data, $q = 12$ corresponds to yearly seasonality. We use *periodicity* for positive recurrence at a lag and *seasonality* for recurrence tied to a calendar cycle.

- When several segments are analyzed jointly, we use a *calendar-lag vocabulary* $V \subseteq [d]$ to place those segments on a common set of candidate lags. Importantly, requiring $\Omega_s \subseteq V$ implies that all segments’ periodicity signatures are drawn from the same set of candidate lags, but every segment can still have its own support and coefficient strengths.

In this section, we reformulate the proposed robust sparse AR framework as an optimization model, in which the aforementioned assumptions can be seamlessly incorporated. We follow several principles in developing the model. First, the model selects a small number of lags, because a dense AR coefficient vector is not a usable periodicity signature. Second, it focuses on nonnegative coefficients for periodicity discovery, because positive lag dependence corresponds to repeated activity, whereas negative lag dependence corresponds to alternation. Third, it tolerates nearby calendar aliases, such as 23/24 or 167/168, because recurring behavior need not align exactly with a prespecified lag. Fourth, it is robust to shocks and outliers. Finally, when comparing locations, periods, or modes, it keeps the lag vocabulary stable so that coefficient changes have the same interpretation across segments.

3.1. Sparse Nonnegative Calendar-Lag Signatures

For a single segment s , the classical AR estimates a dense coefficient vector by least squares. To reformulate AR as a periodicity quantification model, we impose nonnegativity and sparsity:

$$\begin{aligned}
 \min_{\Omega_s, w_s} \quad & \frac{1}{2} \sum_{t \in \mathcal{T}_s} \left(x_{s,t} - \sum_{k \in \Omega_s} w_{s,k} x_{s,t-k} \right)^2 \\
 \text{s.t.} \quad & \Omega_s \subseteq [d], \quad |\Omega_s| \leq \tau, \\
 & w_{s,k} \geq 0, \quad k \in \Omega_s, \\
 & w_{s,k} = 0, \quad k \notin \Omega_s.
 \end{aligned} \tag{1}$$

Equivalently, using the lagged design matrix, the objective becomes $\|\tilde{x}_s - A_s w_s\|_2^2$. The sparsity budget $\tau \in \mathbb{Z}^+$ controls the size of the periodicity signature. The corresponding coefficients quantify how strongly each of these dependencies appears in segment s .

The nonnegativity constraint is central to periodicity discovery. A positive coefficient at lag q means that high values tend to be followed, one calendar cycle later, by high values, and low values by low values. Negative coefficients may improve prediction in some forecasting tasks, but they do not measure positive recurrence. Therefore, throughout the periodicity-discovery analysis we impose $w_{s,k} \geq 0$. In forecasting-only usage, we may relax

this restriction and allow signed coefficients, because prediction and periodicity quantification are different objectives.

3.2. Grouped Calendar Lags

Exact calendar lags are useful anchors, but real time series often distribute recurrence over neighboring lags. Hourly data may place weekly mass at both 167 and 168, or daily mass at both 23 and 24. Treating these as unrelated lags would overstate the number of discovered patterns and understate the strength of the underlying calendar recurrence. We therefore define calendar windows around reference lags.

Let $\mathcal{Q} \subseteq [d]$ be a set of reference lags. For each $q \in \mathcal{Q}$, define $G_q = \{k \in [d] : |k - q| \leq r_q\}$, where $r_q \geq 0$ is a user-specified tolerance. For example, in hourly mobility data, G_{24} and G_{168} may include lags near one day and one week, respectively; in monthly climate data, G_{12} represents a yearly window. The grouped periodicity strength of segment s at reference lag q is $\pi_{s,q} = \sum_{k \in \Omega_s \cap G_q} w_{s,k}$. When $G_q = \{q\}$, this reduces to the exact-lag coefficient $w_{s,q}$. When G_q contains neighboring aliases, $\pi_{s,q}$ measures the total coefficient mass assigned to that calendar recurrence.

Grouped sparsity can also be used as a modeling constraint. Define the selected calendar groups $\mathcal{Q}(\Omega_s) = \{q \in \mathcal{Q} : \Omega_s \cap G_q \neq \emptyset\}$. A grouped model may impose $|\mathcal{Q}(\Omega_s)| \leq B$ and $|\Omega_s \cap G_q| \leq a_q$, $q \in \mathcal{Q}$, where B controls the number of selected calendar groups and a_q controls how many lags may be active within the window around q . These constraints encode calendar knowledge without forcing the model to choose exactly one prespecified lag.

3.3. Robust Periodicity Under Disruptions

Mobility time series contain holidays, storms, airport irregularities, service disruptions, policy interventions, and large shocks. Squared loss can allow a small number of abnormal periods to influence both the selected support and the coefficient magnitudes. To reduce this sensitivity, we replace squared loss with the Huber loss such that

$$\rho_c(\varepsilon) = \begin{cases} \frac{1}{2}\varepsilon^2, & |\varepsilon| \leq c, \\ c|\varepsilon| - \frac{1}{2}c^2, & \text{otherwise,} \end{cases}$$

where $c > 0$ is the robustness threshold. As shown in Figure 2, for small residuals, the Huber loss behaves like squared loss; for large residuals, it grows linearly. Thus, abnormal periods remain in the estimation problem but receive less influence. In this paper, robustness is used to protect the learned periodicity signature from being driven by disruptions.

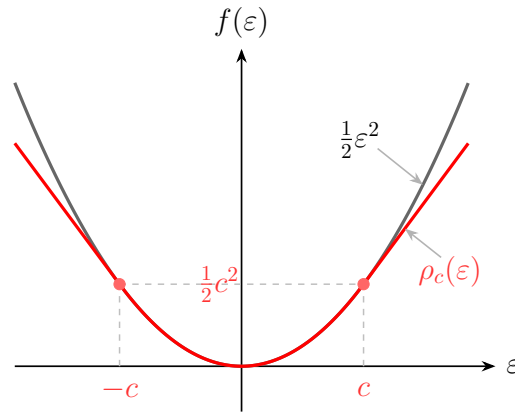


Figure 2 Comparison of quadratic and Huber loss functions. For small residuals within the threshold region $[-c, c]$, both the red (Huber loss) and gray (parabola) curves are quadratic. For residuals outside this region, the red curve is a linear function with constant slope $\pm c$.

3.4. Stability Across Time and Space

When segments are estimated independently, different segments may select different supports. This can obscure the interpretation of coefficient changes. For example, if one year selects lag 168 and another selects lag 167, a direct comparison of $w_{s,168}$ may falsely suggest that weekly periodicity disappeared. Similarly, if two time phases use different lag supports, differences in coefficients may reflect different model vocabularies rather than different mobility rhythms.

We address this issue by stabilizing the lag vocabulary $V \subseteq [d]$ across linked segments. A shared-vocabulary model imposes $\Omega_s \subseteq V$, $|V| \leq \tau$, $s \in \mathcal{S}$. The coefficients $w_{s,k}$ remain segment-specific, so each period or location can have its own periodicity strengths. The constraint only ensures that the strengths are attached to the same candidate lags. A more flexible version allows supports to vary but controls how much they change across linked segments. Let $\mathcal{E}$ be a graph over $\mathcal{S}$. An edge $(s, s') \in \mathcal{E}$ may connect adjacent time periods for the same location, neighboring spatial areas, or different travel modes observed on the same spatial partition. Define the symmetric-difference distance $d_\Delta(\Omega_s, \Omega_{s'}) = |\Omega_s \Delta \Omega_{s'}| = |\Omega_s \cup \Omega_{s'}| - |\Omega_s \cap \Omega_{s'}|$. Controlled support stability can be written as $d_\Delta(\Omega_s, \Omega_{s'}) \leq \kappa_{ss'}$, $(s, s') \in \mathcal{E}$, where $\kappa_{ss'}$ determines how much support variation is allowed. The special case $\kappa_{ss'} = 0$ enforces the same selected support across linked segments. In the main mobility and climate analyses, the most important case is the common-vocabulary constraint, where all segments are placed on the same calendar-lag vocabulary, while coefficient magnitudes are allowed to vary.

3.5. Unified Model and Special Cases

The preceding components can be summarized by a single set-based model:

$$\begin{aligned}
& \min_{V, \{\Omega_s, w_s\}_{s \in \mathcal{S}}} \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}_s} \rho_c \left(x_{s,t} - \sum_{k \in \Omega_s} w_{s,k} x_{s,t-k} \right) \\
& \text{s.t. } \Omega_s \subseteq V \subseteq [d], \quad |V| \leq \tau, \quad s \in \mathcal{S}, \\
& \quad w_{s,k} \geq 0, \quad k \in \Omega_s, \quad s \in \mathcal{S}, \\
& \quad w_{s,k} = 0, \quad k \notin \Omega_s, \quad s \in \mathcal{S}, \\
& \quad |\mathcal{Q}(\Omega_s)| \leq B, \quad |\Omega_s \cap G_q| \leq a_q, \quad q \in \mathcal{Q}, \quad s \in \mathcal{S}, \\
& \quad d_\Delta(\Omega_s, \Omega_{s'}) \leq \kappa_{ss'}, \quad (s, s') \in \mathcal{E}.
\end{aligned} \tag{2}$$

The models used in the paper are special cases of (2). With a single segment, squared loss, nonnegative coefficients, and no grouped or stability constraints, the formulation reduces to sparse nonnegative AR for periodicity discovery. Replacing squared loss with ρ_c gives Huber sparse AR (Huber-SAR). Adding calendar windows G_q gives grouped SAR. Letting $\mathcal{S}$ index time periods and imposing a common vocabulary V gives a temporally stable model for tracking the same lags over time. Letting $\mathcal{S}$ index locations, modes, or location-mode pairs gives a spatially or modally stable model with segment-specific periodicity strengths. Large climate and sea surface temperature panels use the same shared-vocabulary principle, with local coefficient estimation performed after learning the common lag vocabulary.

4. Optimization Algorithms

Section 3 defines periodicity quantification as a set-based optimization problem. This section converts that model into MIO formulations and scalable algorithms. We start with the full exact formulation corresponding to Problem (2), which shows how each modeling requirement—sparsity, nonnegativity, grouped calendar lags, robustness, and stability across segments—is encoded. We then introduce three scalable algorithms: iteratively reweighted least squares (IRLS) for robust losses, decision-variable pruning (DVP) for temporally segmented models, and global-local estimation for large spatial and multimodal panels.

4.1. Exact MIO Formulation

Support and vocabulary variables. Let $z_{s,k} \in \{0, 1\}$ indicate whether lag k is active in the support available to segment s . Let $v_k \in \{0, 1\}$ indicate whether lag k belongs to the common

calendar-lag vocabulary. The support activation constraints are $0 \leq w_{s,k} \leq M_k z_{s,k}$, $z_{s,k} \leq v_k$, $s \in \mathcal{S}$, $k \in [d]$, together with the vocabulary budget $\sum_{k=1}^d v_k \leq \tau$. Thus, segment-level supports are contained in the common vocabulary, but we do not force every segment to use every selected vocabulary lag, and coefficient strengths remain segment-specific.

The constants M_k are activation bounds used to encode the implication $z_{s,k} = 0 \Rightarrow w_{s,k} = 0$. In the periodicity-discovery models with nonnegative coefficients, if stationarity is desired, we can add the constraint $\sum_{k=1}^d w_{s,k} \leq 1 - \epsilon$, $s \in \mathcal{S}$, for a small $\epsilon > 0$. Under non-negativity, this condition is necessary and sufficient for the AR characteristic polynomial $\phi(\zeta) = 1 - \sum_{k=1}^d w_{s,k} \zeta^k$ to have all roots outside the unit circle, see Lemma 1. With this optional constraint, $M_k = 1$ is a valid activation bound. In the main periodicity quantification experiments, we do not impose global AR stationarity unless explicitly stated. When coefficients are normalized as relative periodicity shares, the normalization itself implies $0 \leq w_{s,k} \leq 1$, so we set $M_k = 1$. In unnormalized fits, we obtain M_k from nonnegative least squares or sparse fits, applying a conservative multiplicative factor, and we verify after optimization that no selected coefficient is close to its bound. If a selected coefficient is near M_k , we increase the bound and rerun the model. Thus, exact support-selection claims are conditional on the activation bounds being nonbinding.

LEMMA 1. *Let $w \in \mathbb{R}^d$ with $w_k \geq 0$ for all k , and let $\phi(\zeta) = 1 - \sum_{k=1}^d w_k \zeta^k$. Then all roots of ϕ lie strictly outside the closed unit disk ($|\zeta| > 1$) if and only if $\sum_{k=1}^d w_k < 1$.*

Proof (sufficiency). Suppose $\sum_k w_k < 1$. For any ζ with $|\zeta| \leq 1$,

$$\left| \sum_{k=1}^d w_k \zeta^k \right| \leq \sum_{k=1}^d w_k |\zeta|^k \leq \sum_{k=1}^d w_k < 1,$$

using $w_k \geq 0$ and $|\zeta|^k \leq 1$. Hence $\sum_k w_k \zeta^k \neq 1$, i.e. $\phi(\zeta) \neq 0$, for every $|\zeta| \leq 1$. So every root of ϕ satisfies $|\zeta| > 1$.

Proof (necessity). Suppose $\sum_k w_k \geq 1$. Since $w_k \geq 0$, $g(\zeta) := \sum_k w_k \zeta^k$ is continuous and non-decreasing on $\zeta \in [0, 1]$, with $g(0) = 0$ and $g(1) = \sum_k w_k \geq 1$. By the intermediate value theorem there exists $\zeta^* \in (0, 1]$ with $g(\zeta^*) = 1$, i.e. $\phi(\zeta^*) = 0$ with $|\zeta^*| \leq 1$. Hence not all roots of ϕ lie outside the closed unit disk. ■

Grouped calendar-lag variables. Let $\mathcal{Q}$ be the set of reference calendar lags and $G_q \subseteq [d]$ be the window around $q \in \mathcal{Q}$. For each segment s and calendar group q , let $g_{s,q} \in \{0, 1\}$ indicate whether segment s uses any lag in group G_q . The grouped constraints are: (i) $z_{s,k} \leq g_{s,q}$, $s \in \mathcal{S}$, $q \in \mathcal{Q}$, $k \in G_q$, (ii) $\sum_{k \in G_q} z_{s,k} \leq a_q g_{s,q}$, $s \in \mathcal{S}$, $q \in \mathcal{Q}$, (iii) $g_{s,q} \leq \sum_{k \in G_q} z_{s,k}$, $s \in \mathcal{S}$, $q \in \mathcal{Q}$, and (iv) $\sum_{q \in \mathcal{Q}} g_{s,q} \leq B$, $s \in \mathcal{S}$. The parameter B limits the number of selected calendar groups, while a_q limits the number of lags that may be selected inside group G_q . For example, hourly mobility data may use windows around $q = 24$ and $q = 168$, allowing the model to represent near-daily and near-weekly recurrence without forcing all mass onto one prespecified lag. When the grouping restriction is intended at the vocabulary level rather than at the segment level, the same constraints can be written with v_k and group variables g_q .

Support stability variables. Let $\mathcal{E}$ be a graph over segments. An edge $(s, s') \in \mathcal{E}$ can connect adjacent time periods, neighboring locations, or different travel modes observed on the same spatial partition. To control support changes across linked segments, we introduce $\xi_{ss'k} \in \{0, 1\}$ and impose the following constraints: (i) $\xi_{ss'k} \geq z_{s,k} - z_{s',k}$, $(s, s') \in \mathcal{E}$, $k \in [d]$, (ii) $\xi_{ss'k} \geq z_{s',k} - z_{s,k}$, $(s, s') \in \mathcal{E}$, $k \in [d]$, and (iii) $\sum_{k=1}^d \xi_{ss'k} \leq \kappa_{ss'}$, $(s, s') \in \mathcal{E}$. The case $\kappa_{ss'} = 0$ enforces identical binary supports across linked segments. Larger values allow controlled support variation. In the main applications, we often use the simpler common-vocabulary constraint $z_{s,k} \leq v_k$, because it is enough to ensure that coefficients are interpreted on the same calendar-lag vocabulary.

Robust residual variables. To encode the Huber loss exactly, we introduce slack variables $\alpha_{s,t} \geq 0$ and $\beta_{s,t} \in \mathbb{R}$. For a robustness threshold $c > 0$, the Huber loss can be represented by $\rho_c(\varepsilon) = \min_{\alpha \geq 0, \beta \in \mathbb{R}} \left\{ \frac{1}{2}\beta^2 + c\alpha : -\alpha \leq \varepsilon - \beta \leq \alpha \right\}$. Thus, robust residuals are encoded through $-\alpha_{s,t} \leq \varepsilon_{s,t} - \beta_{s,t} \leq \alpha_{s,t}$, $\alpha_{s,t} \geq 0$, $s \in \mathcal{S}$, $t \in \mathcal{T}_s$.

LEMMA 2 (Exact variational form of the Huber loss). *For every $\varepsilon \in \mathbb{R}$ and $c > 0$, the minimum of $\rho_c(\varepsilon)$ is attained at $\beta^* = \text{clip}(\varepsilon, -c, c)$ and $\alpha^* = |\varepsilon - \beta^*|$.*

Proof. For fixed $\alpha \geq 0$, the feasible set for β is the interval $[\varepsilon - \alpha, \varepsilon + \alpha]$. Since $\frac{1}{2}\beta^2$ is convex, the optimal β is the projection of 0 onto this interval:

$$\beta^*(\alpha) = \text{clip}(\varepsilon, \varepsilon - \alpha, \varepsilon + \alpha) = \begin{cases} \varepsilon - \text{sign}(\varepsilon)\alpha, & \alpha < |\varepsilon|, \\ 0, & \alpha \geq |\varepsilon|, \end{cases}$$

We now minimize the resulting objective over $\alpha > 0$ and consider two cases.

- If $\alpha \geq |\varepsilon|$, then $0 \in [\varepsilon - \alpha, \varepsilon + \alpha]$, so $\beta^* = 0$. The objective value is $c\alpha$, minimized over such α at $\alpha = |\varepsilon|$, yielding $c|\varepsilon|$.
- If $\alpha < |\varepsilon|$, the projection is $\beta^*(\alpha) = \varepsilon - \text{sign}(\varepsilon)\alpha$, and the objective becomes

$$\phi(\alpha) = \frac{1}{2}(\varepsilon - \text{sign}(\varepsilon)\alpha)^2 + c\alpha = \frac{1}{2}(|\varepsilon| - \alpha)^2 + c\alpha.$$

Differentiating with respect to α : $\frac{d\phi(\alpha)}{d\alpha} = -(|\varepsilon| - \alpha) + c$. The unconstrained critical point occurs at $\alpha = |\varepsilon| - c$, which lies in $[0, |\varepsilon|)$ if and only if $|\varepsilon| > c$. If $|\varepsilon| \leq c$, then $\frac{d\phi(\alpha)}{d\alpha} \geq 0$ on the interval, so the minimum occurs at the left boundary $\alpha = 0$, giving $\frac{1}{2}\varepsilon^2$. If $|\varepsilon| > c$, the critical point $\alpha = |\varepsilon| - c$ is feasible and yields $\phi(|\varepsilon| - c) = c|\varepsilon| - \frac{1}{2}c^2$. Evaluating the two regimes and comparing boundary values recovers exactly $\rho_c(\varepsilon)$. The claimed minimizers follow directly from the optimal α and β in each case. ■

Full MIO formulation. Combining these components gives the exact MIO formulation:

$$\begin{aligned} \min_{w,z,v,g,\xi,r,\alpha,\beta} \quad & \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}_s} \left(\frac{1}{2} \beta_{s,t}^2 + c \alpha_{s,t} \right) \\ \text{s.t.} \quad & \varepsilon_{s,t} = x_{s,t} - \sum_{k \in [d]} w_{s,k} x_{s,t-k}, & s \in \mathcal{S}, t \in \mathcal{T}_s, \\ & -\alpha_{s,t} \leq \varepsilon_{s,t} - \beta_{s,t} \leq \alpha_{s,t}, \quad \alpha_{s,t} \geq 0, \quad \beta_{s,t} \in \mathbb{R}, & s \in \mathcal{S}, t \in \mathcal{T}_s, \quad (\text{Huber}) \\ & 0 \leq w_{s,k} \leq M_k z_{s,k}, \quad z_{s,k} \leq v_k, & s \in \mathcal{S}, k \in [d], \\ & \sum_{k \in [d]} v_k \leq \tau, & (\text{global sparsity}) \\ & z_{s,k} \leq g_{s,q}, & s \in \mathcal{S}, q \in \mathcal{Q}, k \in G_q, \\ & g_{s,q} \leq \sum_{k \in G_q} z_{s,k} \leq a_q g_{s,q}, & s \in \mathcal{S}, q \in \mathcal{Q}, \\ & \sum_{q \in \mathcal{Q}} g_{s,q} \leq B, & s \in \mathcal{S}, \quad (\text{group sparsity}) \\ & \xi_{ss'k} \geq z_{s,k} - z_{s',k}, \quad \xi_{ss'k} \geq z_{s',k} - z_{s,k}, & (s, s') \in \mathcal{E}, k \in [d], \\ & \sum_{k \in [d]} \xi_{ss'k} \leq \kappa_{ss'}, & (s, s') \in \mathcal{E}, \quad (\text{support stability}) \\ & z_{s,k}, \xi_{ss'k} \in \{0, 1\}, & s \in \mathcal{S}, (s, s') \in \mathcal{E}, k \in [d], \\ & v_k \in \{0, 1\}, \quad g_{s,q} \in \{0, 1\}, & k \in [d], s \in \mathcal{S}, q \in \mathcal{Q}. \end{aligned} \tag{3}$$

Problem (3) is the exact robust MIO counterpart of the unified model of Problem (2). Several important special cases are obtained by removing constraints or changing the loss.

With a single segment, no grouped constraints, no stability constraints, and squared loss, the formulation reduces to nonnegative SAR. With multiple linked segments and a common vocabulary, the formulation gives temporally or spatially stable SAR. With Huber slack variables, it gives exact robust SAR. The squared-loss version is obtained by removing α, β and minimizing $\frac{1}{2} \sum_{s,t} \varepsilon_{s,t}^2$. Forecasting variants replace $0 \leq w_{s,k} \leq M_k z_{s,k}$ by $-M_k z_{s,k} \leq w_{s,k} \leq M_k z_{s,k}$, because negative coefficients may improve prediction even though they are not interpreted as positive recurrence.

4.2. Iteratively Reweighted Least Squares for Robust SAR

The exact Huber formulation (3) is useful when robust support selection must be solved exactly at moderate scale. However, it introduces one pair of slack variables $(\alpha_{s,t}, \beta_{s,t})$ for every residual. For long time series or many segments, these additional variables can dominate the model size. We therefore use IRLS as a scalable robust approximation.

IRLS uses the fact that the Huber loss behaves like a weighted squared loss: residuals inside the Huber threshold receive weight one, while large residuals receive smaller weights. Given coefficients $w^{(\ell)}$ at iteration $\ell \in \mathbb{Z}^+$, define residuals $\varepsilon_{s,t}^{(\ell)} = x_{s,t} - \sum_{k=1}^d w_{s,k}^{(\ell)} x_{s,t-k}$. The next weights are

$$\vartheta_{s,t}^{(\ell+1)} = \min \left\{ 1, \frac{c}{\max\{|\varepsilon_{s,t}^{(\ell)}|, \eta\}} \right\}, \quad (4)$$

where $\eta > 0$ is a small numerical safeguard. Let $\mathcal{C}_{\text{disc}}$ denote the support, vocabulary, grouped-calendar, and stability constraints from (3), excluding the Huber slack variables.

LEMMA 3 (Half-quadratic majorization of ρ_c). *For any $\bar{\varepsilon} \in \mathbb{R}$, define*

$$\vartheta(\bar{\varepsilon}) = \min \left\{ 1, \frac{c}{\max\{|\bar{\varepsilon}|, \eta\}} \right\},$$

for some small $\eta > 0$, and define the quadratic-in- ε surrogate

$$Q(\varepsilon \mid \bar{\varepsilon}) := \frac{1}{2} \vartheta(\bar{\varepsilon}) \varepsilon^2 + \rho_c(\bar{\varepsilon}) - \frac{1}{2} \vartheta(\bar{\varepsilon}) \bar{\varepsilon}^2.$$

*Then for every $\bar{\varepsilon} \in \mathbb{R}$: (1) **Tangency** $Q(\bar{\varepsilon} \mid \bar{\varepsilon}) = \rho_c(\bar{\varepsilon})$, and (2) **Majorization** $Q(\varepsilon \mid \bar{\varepsilon}) \geq \rho_c(\varepsilon)$ for all $\varepsilon \in \mathbb{R}$.*

Proof. Ignore η (it only guards against literal division at $\bar{\varepsilon} = 0$, where $\vartheta = 1$ regardless). Write $w = \vartheta(\bar{\varepsilon})$.

• Case $|\bar{\varepsilon}| \leq c$ ($w = 1$): here $\rho_c(\bar{\varepsilon}) = \frac{1}{2}\bar{\varepsilon}^2$, so $Q(\varepsilon | \bar{\varepsilon}) = \frac{1}{2}\varepsilon^2 + \frac{1}{2}\bar{\varepsilon}^2 - \frac{1}{2}\bar{\varepsilon}^2 = \frac{1}{2}\varepsilon^2$. Tangency is immediate. For majorization, the identity $\frac{1}{2}\varepsilon^2 - \rho_c(\varepsilon) = \frac{1}{2}\max(|\varepsilon| - c, 0)^2 \geq 0$ holds for every $\varepsilon \in \mathbb{R}$ (check both pieces of ρ_c directly), so $\frac{1}{2}\varepsilon^2 \geq \rho_c(\varepsilon)$ always.

• Case $|\bar{\varepsilon}| = b > c$ ($w = c/b \in (0, 1)$): substituting $\rho_c(\bar{\varepsilon}) = cb - \frac{1}{2}c^2$,

$$Q(\varepsilon | \bar{\varepsilon}) = \frac{1}{2}\frac{c}{b}\varepsilon^2 + \frac{1}{2}cb - \frac{1}{2}c^2.$$

Tangency at $\varepsilon = \bar{\varepsilon}$: $\frac{1}{2}\frac{c}{b}\bar{\varepsilon}^2 + \frac{1}{2}cb - \frac{1}{2}c^2 = cb - \frac{1}{2}c^2 = \rho_c(\bar{\varepsilon})$. For majorization, set $D(\varepsilon) = Q(\varepsilon | \bar{\varepsilon}) - \rho_c(\varepsilon)$:

• On $|\varepsilon| \leq c$: $D(\varepsilon) = \frac{1}{2}\varepsilon^2(c/b - 1) + \frac{1}{2}c(b - c)$, a function of ε^2 with negative coefficient (since $c/b < 1$), so minimized over $|\varepsilon| \leq c$ at $|\varepsilon| = c$, where $D(\pm c) = \frac{c(b - c)^2}{2b} \geq 0$.

• On $|\varepsilon| > c$: writing $u = |\varepsilon|$, $D = \frac{1}{2}\frac{c}{b}u^2 - cu + \frac{1}{2}cb$ is a convex quadratic in u with unconstrained minimizer $u^* = b$ (which lies in the domain $u > c$ since $b > c$), and $D(b) = 0$.

Hence $D(\varepsilon) \geq 0$ for all ε , proving majorization. ■

At iteration $\ell + 1$, IRLS solves the weighted MIO:

$$\begin{aligned} \min_{w, z, v, g, \xi} \quad & \frac{1}{2} \sum_{s \in \mathcal{S}} \sum_{t \in \mathcal{T}_s} \vartheta_{s,t}^{(\ell+1)} \left(x_{s,t} - \sum_{k=1}^d w_{s,k} x_{s,t-k} \right)^2 \\ \text{s.t.} \quad & (w, z, v, g, \xi) \in \mathcal{C}_{\text{disc}}. \end{aligned} \tag{5}$$

This problem has the same structure as the squared-loss sparse AR formulation, with a weighted quadratic objective. After solving (5), residuals are recomputed and the weights are updated using (4). The procedure terminates when the objective, coefficients, or weights change by less than a prescribed tolerance. The IRLS procedure is:

1. Initialize $\vartheta_{s,t}^{(0)} = 1$ for all $s \in \mathcal{S}$ and $t \in \mathcal{T}_s$.
2. Given weights $\vartheta^{(\ell)}$, solve the weighted sparse AR problem (5).
3. Compute residuals $\varepsilon_{s,t}^{(\ell)}$.
4. Update weights using (4).
5. Repeat until convergence.

IRLS is a scalable approximation that preserves the support-selection structure while reducing the robust problem to a sequence of weighted mixed-integer quadratic problems. If each weighted MIQP is solved globally, the alternating half-quadratic objective is non-increasing. Because the model includes integer support variables, the procedure does not generally certify the global optimum of the original Huber MIQP.

PROPOSITION 1 (Monotone descent of IRLS via majorization-minimization).

Let $w^{(0)} \in \mathcal{C}_{\text{disc}}$ be an initial iterate and, for $\ell = 0, 1, 2, \dots$, define

$$\theta_{s,t}^{(\ell+1)} = \vartheta(\varepsilon_{s,t}(w^{(\ell)})), \quad w^{(\ell+1)} = \arg \min_{w \in \mathcal{C}_{\text{disc}}} \sum_{s,t} Q(\varepsilon_{s,t}(w) \mid \varepsilon_{s,t}(w^{(\ell)})),$$

where the second equality holds because the constant term $\rho_c(\bar{\varepsilon}) - \frac{1}{2}\vartheta(\bar{\varepsilon})\bar{\varepsilon}^2$ in each $Q(\cdot \mid \bar{\varepsilon})$ does not depend on w , so minimizing $\sum_{s,t} Q(\varepsilon_{s,t}(w) \mid \varepsilon_{s,t}(w^{(\ell)}))$ over w is identical to solving the weighted least-squares problem (5) with weights $\theta^{(\ell+1)}$. If each weighted MIQP (5) is solved to global optimality, then $\{F(w^{(\ell)})\}_{\ell \geq 0} := \{\sum_{s,t} \rho_c(\varepsilon_{s,t}(w^{(\ell)}))\}$ is non-increasing:

$$F(w^{(\ell+1)}) \leq F(w^{(\ell)}) \quad \text{for all } \ell \geq 0.$$

Proof. Write $Q^{(\ell)}(w) := \sum_{s,t} Q(\varepsilon_{s,t}(w) \mid \varepsilon_{s,t}(w^{(\ell)}))$. Then

$$F(w^{(\ell+1)}) \leq Q^{(\ell)}(w^{(\ell+1)}) \stackrel{w^{(\ell+1)} \text{ minimizes } Q^{(\ell)} \text{ over } w^{(\ell)} \in \mathcal{C}_{\text{disc}}}{\leq} Q^{(\ell)}(w^{(\ell)}) = F(w^{(\ell)}).$$

The middle inequality uses that $w^{(\ell)} \in \mathcal{C}_{\text{disc}}$ is a feasible comparator and $w^{(\ell+1)}$ is a global minimizer of $Q^{(\ell)}$ over $\mathcal{C}_{\text{disc}}$. ■

4.3. Decision-Variable Pruning for Stable Multi-Segment Models

The exact MIO formulation can become expensive when $\mathcal{S}$ indexes many segments. Even if the number of binary calendar-lag vocabulary variables remains d , the number of continuous coefficient variables scales as $|\mathcal{S}|d$. DVP addresses this by reducing the lag set before solving the final coupled MIO: it screens candidate lags segment by segment, takes their union, and then solves the final stable-support MIO on the reduced lag set.

Let τ be the final sparsity budget and let $\tau_0 > \tau$ be a relaxed screening sparsity level. For each segment $s \in \mathcal{S}$, we solve a fast screening problem

$$\begin{aligned} \bar{w}_s \in \arg \min_w \quad & \frac{1}{2} \sum_{t \in \mathcal{T}_s} \omega_{s,t} \left(x_{s,t} - \sum_{k=1}^d w_k x_{s,t-k} \right)^2 \\ \text{s.t.} \quad & w \geq 0, \quad \|w\|_0 \leq \tau_0. \end{aligned} \tag{6}$$

The weights $\omega_{s,t}$ equal one in the squared-loss case. In robust variants, they can be set to the current IRLS weights. Thus, least squares is a screening surrogate. Let $C_s = \text{supp}(\bar{w}_s)$, $C = \bigcup_{s \in \mathcal{S}} C_s$. If grouped calendar windows are used, we expand the candidate set by including the full windows associated with screened lags: $C \leftarrow C \cup \bigcup_{q \in \mathcal{Q}: G_q \cap C \neq \emptyset} G_q$. This prevents the screening step from selecting, e.g., lag 167 while removing its calendar neighbor 168.

The final reduced MIO is then solved on C , not on the full lag set $[d]$. Thus, we solve Problem (3) with constraints $0 \leq w_{s,k} \leq M_k z_{s,k}$, $s \in \mathcal{S}$, $k \in C$, and $w_{s,k} = 0$, $s \in \mathcal{S}$, $k \notin C$. Similarly, grouped calendar constraints and support-stability constraints are added after restricting all lag indices to C . DVP is exact if the optimal vocabulary of the full stable-support MIO is contained in C . Otherwise, it is a controlled approximation whose quality depends on the screening step. The relaxed sparsity level τ_0 is used only during screening; the final reduced MIO uses the target sparsity level τ .

4.4. Global–Local Estimation for Large Panels

DVP reduces the lag dimension but still solves a coupled MIO with segment-specific coefficient vectors. This remains expensive when $\mathcal{S}$ contains many locations, modes, location-mode pairs, or climate grid-cell periods. In those settings, we use a global–local decomposition. The method first learns one admissible calendar-lag vocabulary from the full panel and then estimates local coefficient strengths independently under that vocabulary.

Let $\mathcal{V} \subseteq \{0, 1\}^d$ denote the set of admissible vocabulary vectors. This set encodes the global discrete restrictions from the model, such as sparsity, grouped calendar windows, and any other vocabulary-level restrictions used in the application. For $v \in \mathcal{V}$, define $V(v) = \{k \in [d] : v_k = 1\}$. Let $\mathcal{W}(v)$ denote the global coefficient feasible set associated with vocabulary $V(v)$, and let $\mathcal{W}_s(v)$ denote the corresponding local feasible set for segment s . For a coefficient vector w , we formulate $L_s(w) = \sum_{t \in \mathcal{T}_s} f\left(x_{s,t} - \sum_{k=1}^d w_k x_{s,t-k}\right)$, where $f(\varepsilon) = \varepsilon^2/2$ for squared loss and $f(\varepsilon) = \rho_c(\varepsilon)$ for Huber loss. The global stage solves the common-vocabulary surrogate

$$(\hat{\bar{w}}, \hat{v}) \in \arg \min_{\bar{w}, v} \sum_{s \in \mathcal{S}} \lambda_s L_s(\bar{w}) \quad \text{s.t.} \quad v \in \mathcal{V}, \quad \bar{w} \in \mathcal{W}(v), \quad (7)$$

where $\lambda_s \geq 0$ are optional segment weights. The vector $\hat{v}$ defines the common calendar-lag vocabulary $\hat{V} = V(\hat{v})$. The local stage estimates segment-specific coefficients independently:

$$\hat{w}_s \in \arg \min_{w \in \mathcal{W}_s(\hat{v})} L_s(w), \quad s \in \mathcal{S}. \quad (8)$$

Thus, all segments share the same vocabulary $\hat{V}$, but each has its own coefficient strengths.

The decomposition is compatible with the structural constraints in the unified model whenever those constraints can be represented either as vocabulary-level restrictions in $\mathcal{V}$ or as local coefficient restrictions in $\mathcal{W}_s(\hat{v})$. It is not intended to solve the most general

individualized support-variation model with arbitrary segment-specific binary supports. If such support variation is central, one should use the full MIO or the DVP-reduced coupled MIO. In large-panel experiments, the shared-vocabulary version keeps coefficient interpretation aligned across many segments while avoiding millions of local binary variables.

For computation, the least squares and IRLS cases are especially convenient. At IRLS iteration $\ell \in \mathbb{Z}^+$, define the weighted loss

$$L_s^{(\ell)}(w) = \frac{1}{2} \left\| (W_s^{(\ell)})^{1/2} (\tilde{x}_s - A_s w) \right\|_2^2,$$

where $W_s^{(\ell)} = \text{diag}(\{\vartheta_{s,t}\}_{t \in \mathcal{T}_s}) \in \mathbb{R}^{|\mathcal{T}_s| \times |\mathcal{T}_s|}$ is diagonal. Squared loss is recovered by setting $W_s^{(\ell)} = I$ as an identity matrix. The global objective can then be aggregated through $P^{(\ell)} = \sum_{s \in \mathcal{S}} \lambda_s A_s^\top W_s^{(\ell)} A_s$, $q^{(\ell)} = \sum_{s \in \mathcal{S}} \lambda_s A_s^\top W_s^{(\ell)} \tilde{x}_s$. The global stage becomes

$$(\hat{w}^{(\ell)}, \hat{v}^{(\ell)}) \in \arg \min_{\bar{w}, v} \frac{1}{2} \bar{w}^\top P^{(\ell)} \bar{w} - (q^{(\ell)})^\top \bar{w} \quad \text{s.t.} \quad v \in \mathcal{V}, \quad \bar{w} \in \mathcal{W}(v). \quad (9)$$

Given $\hat{v}^{(\ell)}$, local coefficients are updated by

$$\hat{w}_s^{(\ell)} \in \arg \min_{w \in \mathcal{W}_s(\hat{v}^{(\ell)})} L_s^{(\ell)}(w), \quad s \in \mathcal{S}. \quad (10)$$

For robust global–local estimation, residuals from the local fits are used to update the IRLS weights,

$$\vartheta_{s,t}^{(\ell+1)} = \min \left\{ 1, \frac{c}{\max\{|\varepsilon_{s,t}^{(\ell)}|, \eta\}} \right\},$$

and the weighted global and local stages are repeated until convergence. When all weights are one, the procedure reduces to the squared-loss global–local method.

The global support problem in (7) or (9) is solved exactly for the chosen global surrogate and the subsequent local problems estimate segment-specific strengths under the learned vocabulary; however, the two-stage procedure does not provide a global optimum certificate for the fully individualized MIO formulation.

5. Case Studies and Results

The empirical section is organized as a sequence of nested case studies. The introductory Manhattan ridehailing example in Figure 1 validates the basic coefficient interpretation: a large coefficient at the weekly lag corresponds to visible x_t -versus- x_{t-168} alignment and stable weekly profile overlays. We therefore begin here with airport ridehailing trips, a

simple temporal setting in which we can test exact support selection, DVP, and robust loss under disrupted dynamics. We then move from the airport to citywide ridehailing trips ([New York City Taxi and Limousine Commission 2024](#)) and subway ridership ([Metropolitan Transportation Authority 2023](#)) in Manhattan, where the same weekly-lag vocabulary tracks disruption and recovery across neighborhoods and years. Across all mobility data experiments, hourly time series data are modeled with order $d = 168$, corresponding to one week of the hourly resolution. Unless otherwise stated, periodicity discovery experiments use nonnegative coefficients and report weekly periodicity through the coefficients or grouped coefficient mass associated with $k = 168$. Next, we use Manhattan ridehailing trips to test whether the same calendar-lag vocabulary can compare recurrence across different phases of the congestion pricing policy that took effect in January 2025. Finally, we use North American climate and global sea surface temperature as external stress tests for scalability and generality, because both contain known yearly seasonality over massive spatial panels.

5.1. Airport Ridehailing: Sparse Support Selection and Robustness

The first case study focuses on ridehailing trips at John F. Kennedy International Airport (JFK). Airport mobility provides a useful intermediate setting: it is simpler than a citywide panel, but it is still affected by daily schedules, weekly travel rhythms, and major disruptions. Thus, this case study addresses two questions. First, does exact support selection matter when the selected lags are interpretable? Second, does robust estimation improve performance under disrupted mobility dynamics?

5.1.1. Data and temporal structure. We aggregate ridehailing pickup trips at JFK into hourly time series. Figure 3 shows the 2019–2021 pickup trip time series as daily heatmaps. The 2019 panel displays recurring daily and weekly structure, whereas the 2020 panel shows the sharp disruption induced by the COVID-19 period. The 2021 panel shows partial recovery, but the temporal pattern does not simply return to the 2019 baseline. This motivates the use of robust SAR models: sparsity identifies the lags that explain recurrence, while robustness prevents abnormal periods from dominating the fitted coefficients.

5.1.2. Exact support selection. We compare three methods for sparse support selection: nonnegative subspace pursuit (NNSP), MIQP with decision-variable pruning (MIQP-DVP), and the full MIQP. The comparison uses JFK pickup trip time series from 2019

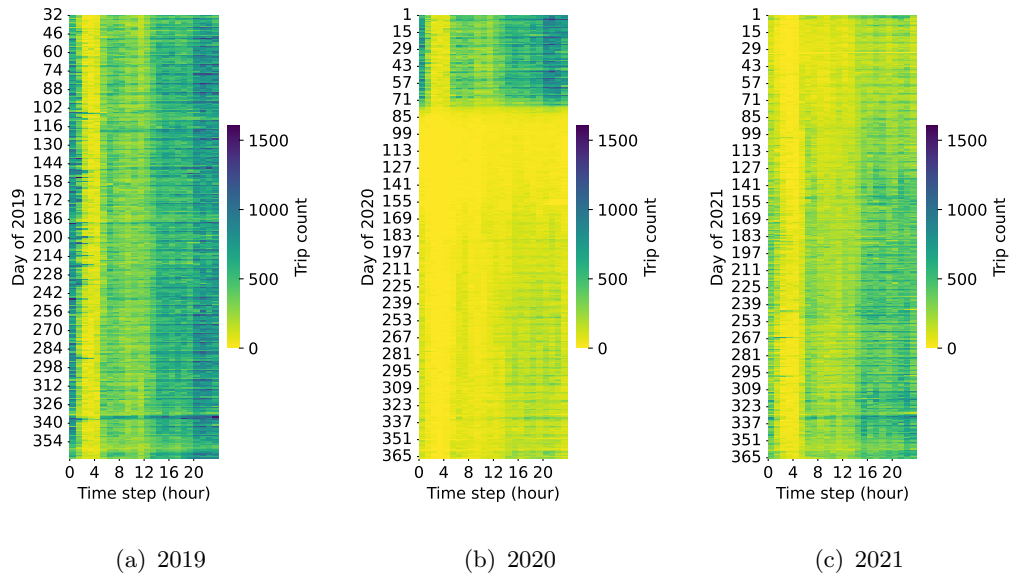


Figure 3 Hourly ridehailing pickup trips at JFK, shown as day-by-hour heatmaps for (a) 2019, (b) 2020, and (c) 2021. Each row is a day of the year and each column is an hour of the day.

through 2023 at sparsity levels $\tau = 4$ and $\tau = 6$. For MIQP-DVP, we test screening levels $\tau_0 = 5$ and $\tau_0 = 10$.

Table 1 shows that full MIQP achieves the best objective values as expected. More importantly, MIQP-DVP with $\tau_0 = 10$ matches the MIQP objective values in every reported year and sparsity level, while reducing average runtime from about 221–223 seconds to less than one second. The more aggressive pruning level $\tau_0 = 5$ can lose the optimal support, particularly in years where the time series is more disrupted. These results support the algorithmic design in Section 4.3: DVP is useful because it screens the lag set aggressively enough to accelerate MIQP, but it must retain a relaxed support large enough to avoid deleting important calendar lags.

Table 1 Objective function of SAR on the ridehailing pickup trip time series at JFK. The solution algorithms include NNSP, MIQP-DVP, and MIQP. The unit of objective function values is $\times 10^7$. Note that the lowest objective function values are emphasized in bold fonts. The last two columns present the average computational times (in seconds).

Method	2019		2020		2021		2022		2023		Cost	
	$\tau = 4$	$\tau = 6$	$\tau = 4$	$\tau = 6$	$\tau = 4$	$\tau = 6$	$\tau = 4$	$\tau = 6$	$\tau = 4$	$\tau = 6$	$\tau = 4$	$\tau = 6$
NNSP	8.48	8.41	2.12	2.03	3.11	3.06	6.85	6.69	8.59	8.39	0.03 s	0.04 s
MIQP-DVP ($\tau_0 = 5$)	8.48	-	2.12	-	3.11	-	6.76	-	8.45	-	0.32 s	-
MIQP-DVP ($\tau_0 = 10$)	8.24	8.07	1.90	1.86	3.06	2.97	6.49	6.34	8.14	7.95	0.67 s	0.66 s
MIQP	8.24	8.07	1.90	1.86	3.06	2.97	6.49	6.34	8.14	7.95	221 s	223 s

The interpretation is also substantive. Because periodicity is read from selected lags, support-selection errors are not merely numerical errors; they change the explanation of the time series. An exact or near-exact support such as one containing local, daily, and weekly lags gives a coherent periodicity signature, whereas a heuristic support with spurious lags can obscure the calendar structure.

5.1.3. Robust forecasting under disruption. We next test whether robust losses add value beyond sparsity. The forecasting task trains on the 2019 JFK ridehailing pickup trip time series and predicts the 2020 pickup trip time series with a one-step-ahead prediction window. Because this is a forecasting experiment rather than a periodicity-discovery experiment, we allow signed coefficients by replacing the nonnegativity constraint with signed big- M bounds. This distinction is important: signed coefficients may improve prediction, but only nonnegative coefficients are interpreted as positive recurrence in the periodicity analyses.

Figure 4 reports prediction error as a function of the AR order d . Dense AR becomes unstable at larger orders, whereas sparse AR remains more stable because the support budget prevents the model from spreading mass across many weak lags. Huber-SAR improves further by downweighting large residuals associated with abnormal periods. The comparison between ℓ_2 -SAR and Huber-SAR highlights the significance of Huber losses for handling noisy time series with outliers.

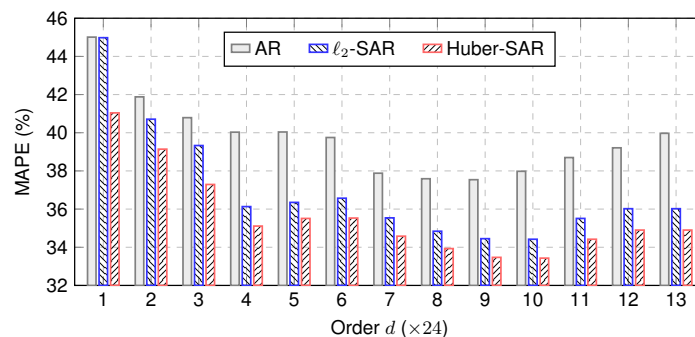


Figure 4 Prediction error (mean absolute percentage error, MAPE) of AR, ℓ_2 -SAR, and Huber-SAR on the ridehailing pickup trip time series at JFK, trained on 2019 data and evaluated on 2020. The AR order $d \in \{24, 48, \dots, 312\}$ is varied to test each model’s stability across lag horizons. Dense AR degrades at larger orders, while sparsity and Huber’s robust loss reduce prediction error consistently across orders.

Table 2 gives the same comparison at order $d = 168$. The best Huber-SAR configuration reaches 34.58% MAPE at $\tau = 10$ and $\delta = 1$, compared with 35.54% for squared-loss SAR at

the same sparsity level and 36.73% for Huber-AR without sparsity. Thus, robustness and sparsity contribute different benefits: sparsity controls the lag structure, while the Huber loss reduces the influence of disrupted observations.

Table 2 Prediction performance (in MAPE) of ℓ_2 -SAR and Huber-SAR models on the ridehailing pickup trip time series at JFK, with AR order $d = 168$. Columns for ℓ_2 -SAR compare fixed and time-varying coefficients; columns for Huber-SAR vary the robustness threshold c . ARIMA(1,2,1) is included as a non-sparse benchmark.

Sparsity	ℓ_2 -SAR		Huber-SAR					ARIMA(1,2,1)
	Fixed w	Time-varying	$c = 1$	$c = 5$	$c = 10$	$c = 15$	$c = 20$	
$\tau = d$	37.88%	37.76%	36.92%	36.86%	36.81%	36.75%	36.73%	58.20%
$\tau = 15$	35.73%	36.32%	34.70%	34.67%	34.64%	34.63%	34.62%	-
$\tau = 10$	35.54%	35.38%	34.58%	34.60%	34.61%	34.62%	34.63%	-
$\tau = 5$	38.01%	37.39%	36.98%	36.92%	36.91%	36.92%	36.93%	-

5.2. Manhattan Ridehailing and Subway Ridership: Disruption and Recovery

The experiment on the JFK ridehailing pickup trip time series isolates exact support selection and robustness in a relatively simple setting. We now move to citywide mobility systems, where the question is whether a common weekly-lag vocabulary can reveal disruption and recovery from COVID-19 across urban areas. We analyze the ridehailing dropoff trips from 2019 to 2025 ([New York City Taxi and Limousine Commission 2024](#)) and the subway ridership from 2020 to 2024 ([Metropolitan Transportation Authority 2023](#)) in Manhattan. For both travel modes, we set $\tau = 6$. The optimized supports of ridehailing dropoff trips and subway ridership are $\Omega = \{1, 23, 24, 143, 167, 168\}$ and $\Omega = \{1, 23, 24, 144, 167, 168\}$, respectively. The supports include local dependence, near-daily lags, and near-weekly lags. We report weekly periodicity using the coefficients at weekly-lag $k = 168$.

Figure 5 shows the main results of Manhattan mobility with ridehailing and subway modes. Of the ridehailing results, weekly periodicity drops from 0.215 in 2019 to 0.152 in 2020 because of disruption related to COVID-19, then rises to 0.201 and 0.208 from 2021 to 2022. In 2023, the weekly periodicity of ridehailing dropoff trips becomes 0.178, then recovers to 0.190 in 2025. Of the subway ridership results, weekly periodicity increases from 0.270 in 2020 to 0.362 in 2021, then remains at roughly the same level, between 0.44 and 0.45, in 2022, 2023, and 2024. The heatmaps show that the disruption is spatially heterogeneous: the ridehailing dropoff trips and the subway ridership in the south areas of Manhattan are generally less periodic than in the north areas of Manhattan, while the

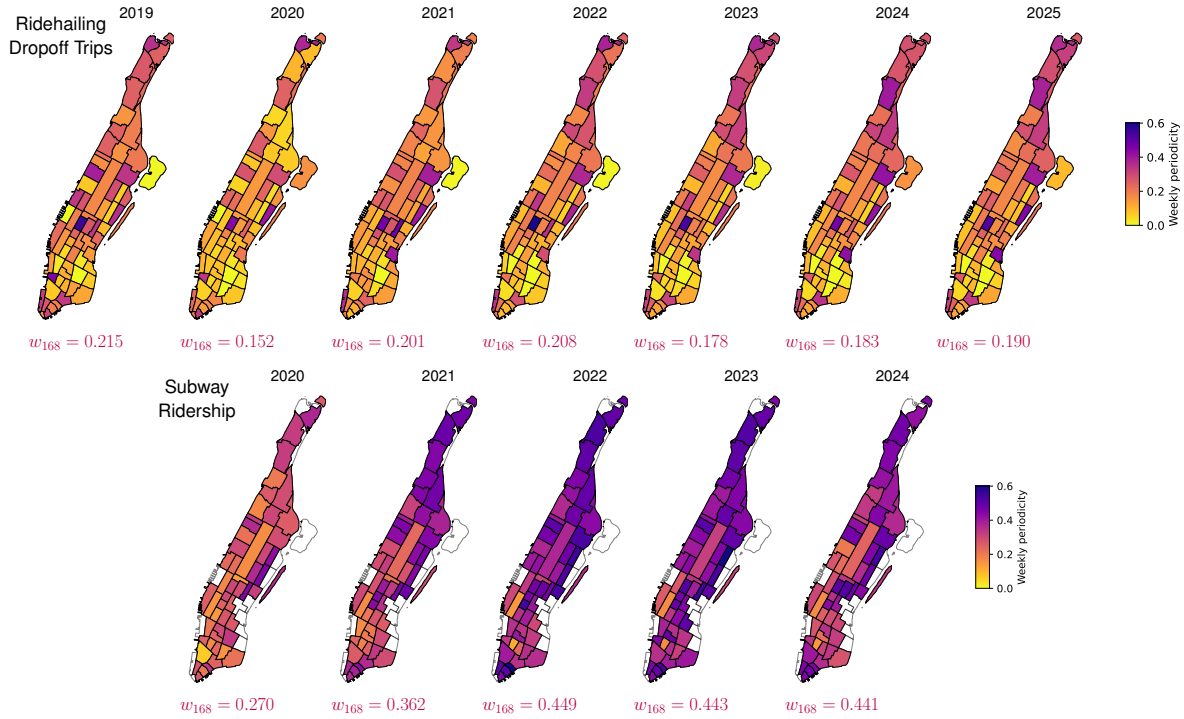


Figure 5 Weekly periodicity of hourly ridehailing dropoff trips (2019–2025) and subway ridership (2020–2024) across 66 areas in Manhattan. Ridehailing periodicity drops sharply in 2020 due to COVID-19 disruption and only partially recovers through 2025, while subway periodicity rises through 2021 and remains consistently higher thereafter. White areas indicate locations without subway service.

travel purposes such as commuting, shopping, and tourism in the south areas of Manhattan are more complicated than in the north areas. Beyond trip volume for measuring activity scale, weekly periodicity is an important metric for measuring regularity of activity. The mobility recovery experiment extends the spatial interpretation of weekly periodicity over time and across two different travel modes.

5.3. Manhattan Ridehailing: Examining the Effects of Congestion Pricing Policy

The previous experiment compares the same travel mode across space and time. We next test whether a common lag vocabulary for hourly ridehailing dropoff trips allows coefficients to be comparable across different phases of policy intervention in urban areas. We consider two 8-week phases after the congestion pricing policy that took effect on January 5, 2025 in southern Manhattan. In our setting, the warm-up phase runs from January 5 to March 1, 2025, while the full implementation phase runs from March 2 to April 26, 2025. The resulting panel contains 66 spatial areas and two phases. The support is optimized as $\Omega = \{1, 23, 24, 143, 167, 168\}$ by setting $\tau = 6$, so that weekly periodicity is captured by the lag $k = 168$. The common support allows the weekly coefficient to be interpreted

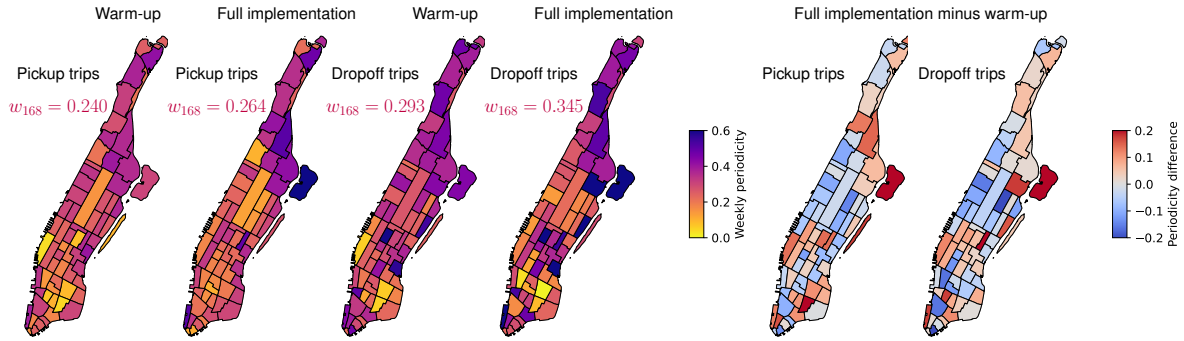


Figure 6 Weekly periodicity of hourly ridehailing pickup and dropoff trips across 66 areas in Manhattan in 2025, before and after the full implementation of the congestion pricing policy that took effect on January 5, 2025. The warm-up phase runs from January 5 to March 1, 2025, and the full implementation phase runs from March 2 to April 26, 2025. Weekly periodicity increases for both pickup and dropoff trips, with the largest gains concentrated in southern Manhattan, inside the congestion relief zone.

consistently across spatial areas and phases, while the coefficient magnitudes remain area- and phase-specific.

Figure 6 shows clear differences in weekly periodicity between the warm-up and full implementation phases, allowing one to discover the effects of the congestion pricing policy in Manhattan. The weekly periodicity of pickup trips is 0.240 during the warm-up phase and then increases to 0.264 during the full implementation phase, consistent with travelers adjusting to the policy. The weekly periodicity of dropoff trips is stronger than that of pickup trips, but it follows a similar pattern, increasing from 0.293 to 0.345. Across 66 spatial areas, the weekly periodicity within southern Manhattan (i.e., congestion relief zone) generally increases, while the weekly periodicity of central Manhattan outside the congestion relief zone becomes much smaller. This analysis of the effects of the congestion pricing policy in Manhattan justifies the spatial stability layer of the model. By using a common calendar-lag vocabulary, the model makes the comparison across spatial areas and time phases interpretable.

5.4. North American Climate: Scalability and Yearly Seasonality

The mobility case studies evaluate coefficient interpretation, support quality, robustness, temporal recovery, spatial heterogeneity, and modal comparison. We now use climate data as an external stress test. The goal is not to make climate-science causal claims, but to test whether the same global-support/local-estimation principle scales to massive spatial panels and recovers interpretable yearly seasonality. We use monthly Daymet climate variables

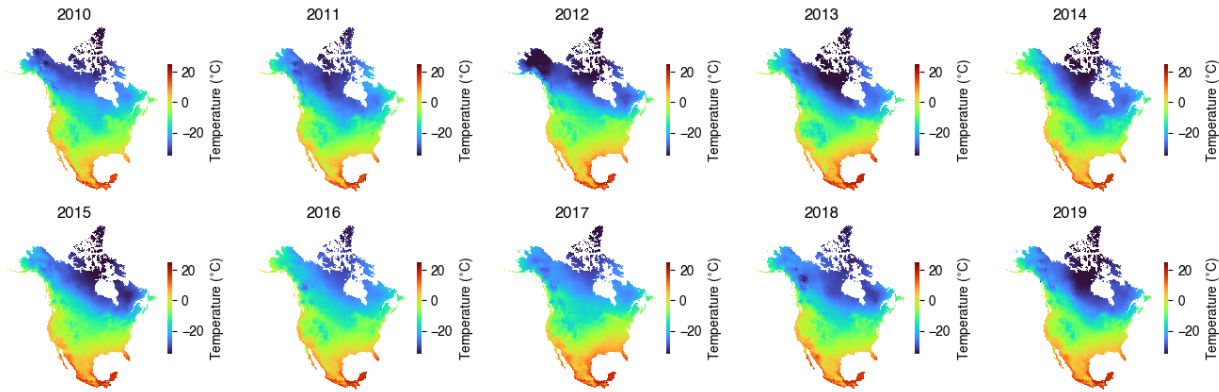


Figure 7 Monthly aggregated minimum temperature across North America of January from 2010 to 2019. The color scale refers to the temperature in degrees Celsius, with blue and red tones indicating lower and higher temperatures, respectively.

from 1980 to 2019 over North America: minimum temperature, maximum temperature, and precipitation (Thornton et al. 2022).

Figure 7 shows January minimum temperature fields from 2010 to 2019. It illustrates the spatial scale of the problem and the strong seasonal structure that motivates a yearly-lag model. At $10\text{ km} \times 10\text{ km}$ resolution, the panel contains more than 800,000 time series. The model order is $d = 12$, so $k = 12$ corresponds to yearly seasonality. Because the monthly climate variable time series are smoother than disrupted mobility time series, we use the squared-loss version of the model, corresponding to the limit $c \rightarrow \infty$ in the Huber loss. The support is optimized as $\Omega = \{1, 11, 12\}$ by setting $\tau = 3$, capturing local monthly cycle and near-yearly/yearly seasonality.

Figure 8 validates this support by comparing it with independent sparse fits across more than 200,000 grid-cell series at $10\text{ km} \times 10\text{ km}$ resolution. Even without enforcing a shared support, the most frequently selected lags are 1, 11, and 12, and higher sparsity levels add nearby lags rather than replacing this structure. This verifies the use of a common global vocabulary for the climate panel.

For the scalability of the global-local procedure, the model fits 3,343,628 time series at $5\text{ km} \times 5\text{ km}$ resolution in 48.55 seconds, 816,612 time series at $10\text{ km} \times 10\text{ km}$ resolution in 11.77 seconds, and 196,720 time series at $20\text{ km} \times 20\text{ km}$ resolution in 2.88 seconds. These runtimes demonstrate why the decomposition is necessary: it avoids solving a mixed-integer support-selection problem separately for every grid cell while preserving local coefficient estimates.

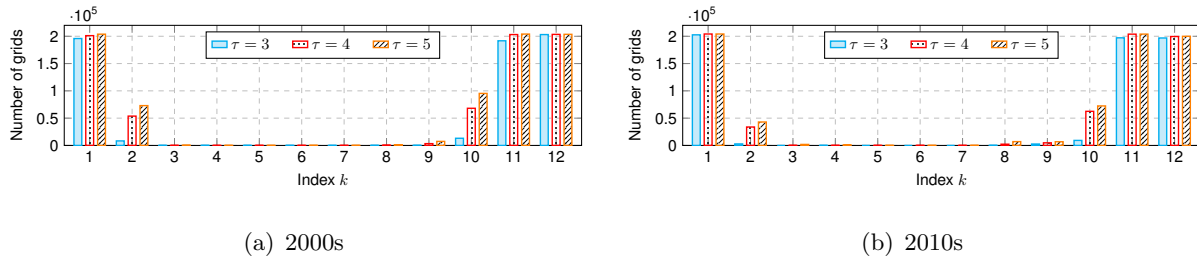


Figure 8 Support sets of independent SAR models with different sparsity levels tested on the minimum temperature data in 2000s and 2010s across 204,153 grids.

Figure 9 reports decadal maps of yearly seasonality strength for minimum temperature, maximum temperature, and precipitation. Temperature seasonality is strongest in high-latitude regions such as Canada and Alaska and weaker or more spatially variable in southern regions. Precipitation exhibits more localized spatial heterogeneity. The 2000s and 2010s show visibly different spatial patterns from the 1980s and 1990s, especially for temperature variables. These results show that the same coefficient-based periodicity measure can be used to map yearly seasonality across variables, space, and decades.

5.5. Sea Surface Temperature: Global Generalization

The final case study tests whether the same framework produces interpretable seasonality maps on a global ocean panel. We use monthly NOAA Optimum Interpolation Sea Surface Temperature data from January 1982 to December 2019 (Huang et al. 2020). The data are observed on a $0.25^\circ \times 0.25^\circ$ grid with 720×1440 cells, yielding 1,036,800 spatial locations. In our model, we set $d = 12$ and use the coefficient at lag 12 as the yearly-seasonality strength with sparsity level $\tau = 3$. The support is optimized as $\Omega = \{1, 11, 12\}$, including the cycle of yearly seasonality.

Figure 10 visualizes yearly seasonality strength by decade across the full 1,036,800-location global sea surface temperature panel. Regions near Eastern and Central Asia, North America, Europe, and North Africa exhibit stronger yearly seasonality. Equatorial regions, including the tropical Pacific, exhibit weaker yearly coefficients, indicating less regular annual recurrence in the monthly sea surface temperature series. As with the North American climate panel in Section 5.4, we interpret these patterns as descriptive seasonality maps rather than as causal attribution to specific oceanographic mechanisms; the goal here is to confirm that the same coefficient-based periodicity measure generalizes from a regional land panel to a global ocean panel of comparable scale.

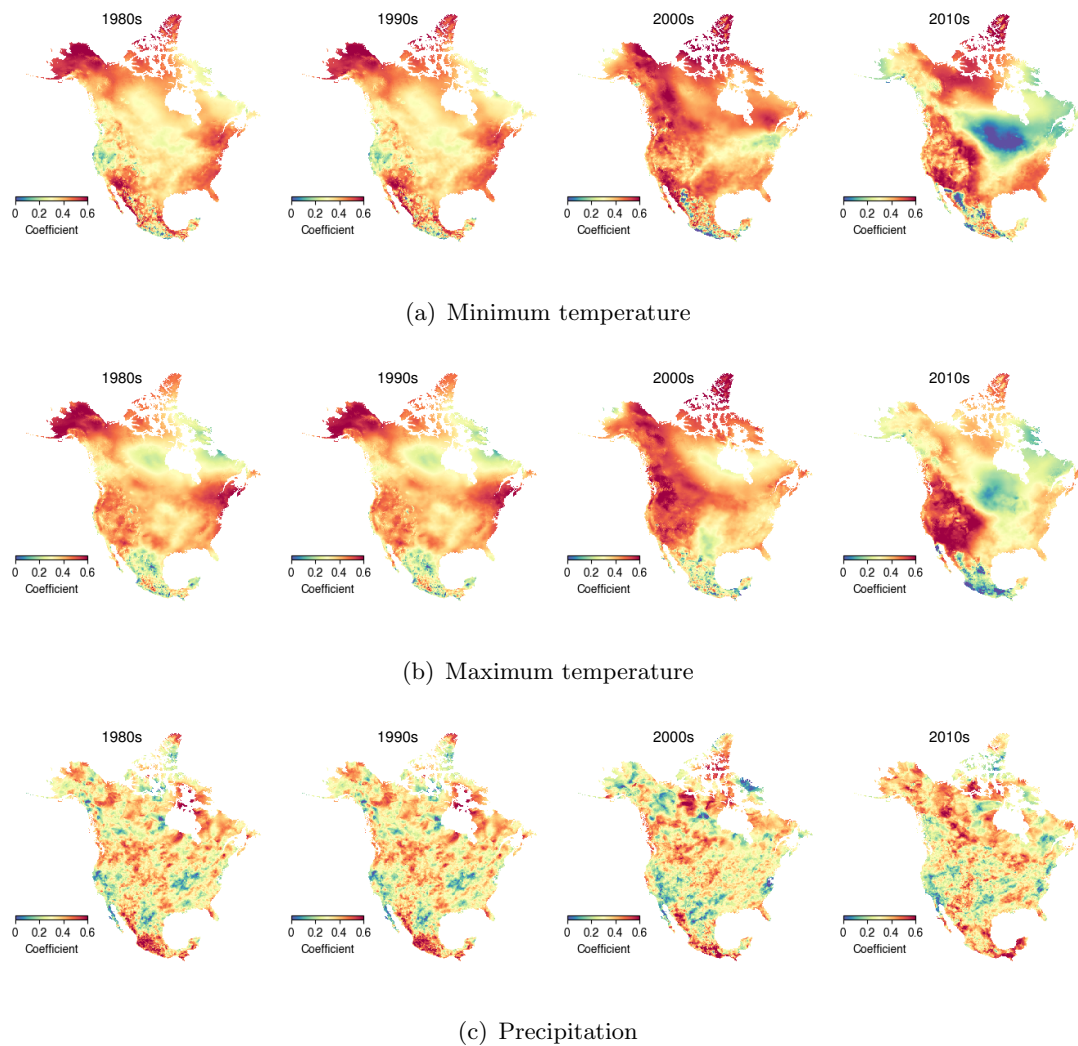


Figure 9 Spatial patterns of the strengths of yearly seasonality quantified by SAR on the North America climate data across the past four decades.

Figure 11 highlights regional changes near Canada and the Arctic Ocean, a region of particular interest because it has experienced some of the fastest climate change on record. The yearly coefficient weakens over time in parts of this region, most visibly across northern Canada, where the strong red band of high seasonality strength in the 1980s panel is visibly reduced by the 2010s. This suggests that the strength of annual recurrence is not spatially or temporally fixed. This final case study reinforces the generality of the framework: the same sparse AR coefficient used to quantify weekly mobility recurrence can also quantify yearly seasonality in a global climate panel, and its decadal changes are consistent with independently documented Arctic warming trends.

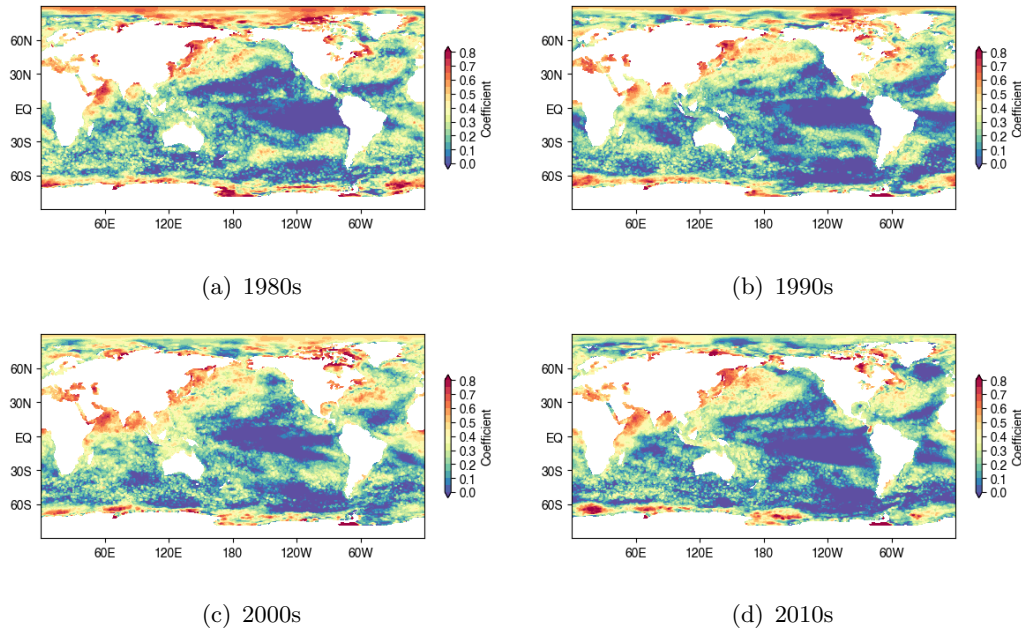


Figure 10 Spatial patterns of the strengths of yearly seasonality quantified by SAR on the sea surface temperature data in the past four decades.

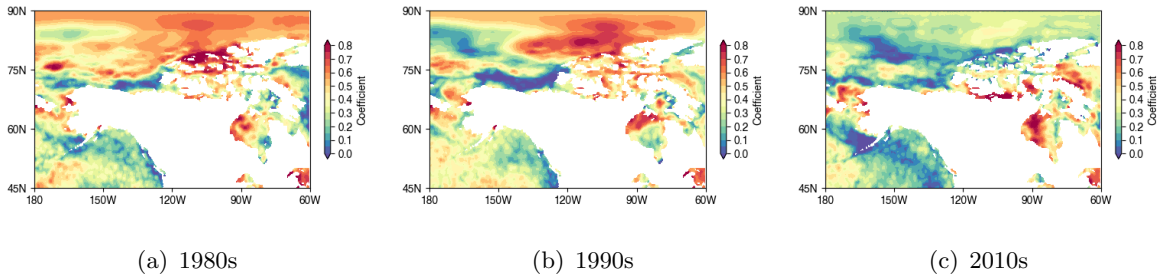


Figure 11 Spatial patterns of the strengths of yearly seasonality on the sea surface temperature data with highlighted areas around Canada.

6. Conclusion and Discussion

This paper develops an optimization framework for interpretable time series autoregression that quantifies periodicity in large collections of related time series. The framework differs from classical autoregression along four dimensions motivated directly by the research question: sparse nonnegative coefficients isolate a small periodicity signature rather than a dense, sign-mixed set of lags; grouped calendar-lag constraints absorb near-calendar aliases so that recurrence is not artificially split across neighboring lags; Huber losses limit the influence of disruptions on the estimated signature; and a shared calendar-lag vocabulary keeps coefficient comparisons meaningful across time and space. We formulated these requirements as a single MIO model and then developed a sequence of algorithms,

including IRLS, DVP, and global-support/local-estimation decomposition, which make the model tractable from single-segment problems up to panels of several million series.

Summary of findings. The case studies were organized to test the framework’s claims in increasing order of scale and complexity. At JFK, exact support selection mattered: MIQP-DVP recovered the same objective values as the full MIQP while cutting runtime from several minutes to under a second, and Huber-SAR consistently outperformed both dense AR and squared-loss SAR in one-step-ahead forecasting during the COVID-19 disruption, showing that sparsity and robustness contribute distinct, complementary benefits rather than substituting for one another. In Manhattan, imposing a common weekly-lag vocabulary made it possible to track the disruption and recovery of ridehailing and subway regularity across neighborhoods and years, and to compare periodicity before and after the congestion pricing policy. These comparisons would not be well defined if each segment were allowed to select its own idiosyncratic support. The North American climate and global sea surface temperature studies stress-tested the global-local decomposition at a scale unrelated to urban mobility, fitting on the order of 800,000 to more than three million series in seconds to tens of seconds while recovering a stable, physically sensible $\{1, 11, 12\}$ monthly-yearly support and interpretable decadal seasonality maps, including a visible weakening of yearly recurrence over parts of the rapidly warming Arctic region. Taken together, these results support the paper’s central claim: a coefficient-based periodicity measure, estimated under sparsity, nonnegativity, calendar grouping, robustness, and support stability, is simultaneously interpretable, accurate, and scalable.

Limitations. Several limitations qualify the scope of these conclusions. First, DVP and the global-local decomposition are both approximations to the fully individualized, jointly coupled MIO: DVP is exact only when the true optimal vocabulary is contained in the screened candidate set, and the global-local method does not certify a global optimum for the disaggregated problem, even though it performs well empirically. Second, robustness is handled through a Huber loss whose purpose is to prevent disruptions from dominating the estimated signature, not to identify, date, or causally attribute change points; the framework should not be read as a change-point detector or an outlier-classification procedure. Third, the empirical studies are descriptive rather than causal. This is explicit in the climate and sea surface temperature analyses, and it also applies to the congestion pricing comparison, where periodicity shifts are associated with, but not proven to be caused by,

the policy intervention. Finally, the choice of calendar reference lags Q and window radius r_q is guided by domain knowledge (e.g., 24 and 168 for hourly mobility data, 12 for monthly climate data); the framework does not currently learn these reference points from data, so it presumes at least approximate prior knowledge of the relevant periodic cycles.

Extensions. These limitations point to natural directions for future work. One is to develop tighter or data-driven procedures for setting activation bounds and pruning thresholds, potentially with finite-sample guarantees on when DVP and global-local estimation coincide with the fully coupled MIO solution. A second is to relax the assumption of known reference lags, allowing the model to discover calendar-relevant cycles directly from the data rather than only from a prespecified candidate set. A third is to strengthen the connection between the robust periodicity signature and formal change-point or regime-switching models, so that disruption and recovery can be localized in time as well as downweighted in estimation. A fourth is to extend the support-stability constraints beyond the common-vocabulary special case to the more general graph-based formulation, applying it to settings where segments are only partially linked. We view the present framework as a foundation for these extensions: a single, unifying optimization model whose special cases already scale from one time series to systems of millions.

Acknowledgments

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